

SUJAL DHAKAL

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EDUCATION

Georgia Institute of Technology

Expected May 2027

Bachelor of Science in Computer Science

Atlanta, GA

- **GPA:** 3.9
- **Concentrations:** Artificial Intelligence & Human-Centered Systems
- **Related Coursework:** Data Structures & Algorithms, Databases & SQL, Algorithm Design, Computer Architecture & Programming, User Interface Design, Software Design & UML, Discrete Mathematics, Linear Algebra

EXPERIENCE

Bits of Good — Hack4Impact Chapter

Jan. 2025 – Present

Engineering Manager

Atlanta, GA

- Orchestrated development of a medical distribution platform serving **56 facilities** and **\$50M+** in annual supplies.
- Re-architected a Next.js + React codebase, refactoring **170+ files** into a modular controller–service–repository architecture and reusable hooks/providers, enabling rapid feature delivery for an 8-person team.
- Reduced manual effort by **80%** by automating distribution workflows via GPT-5, Azure, and RAG pipelines.
- Authored **60+ GitHub issues**, reviewed **400+ commits**, and resolved conflicts while mentoring developers.

Georgia Institute of Technology

Aug. 2024 – Present

Undergraduate Researcher

Atlanta, GA

- Preprocessed **2.4 million** biometric data points from Apple HealthKit using Pandas, cleaned into time-series data.
- Achieved **91% accuracy** with a semi-supervised LSTM model in PyTorch, predicting health abnormalities.
- Developed Swift app that streams health data via Firebase to the model for monitoring, alerts, and automations.

Technique Newspaper

Nov. 2024 – Sept. 2025

Software Developer

Atlanta, GA

- **Cut page load by 8s** by replacing WordPress front-end with static-rendered Next.js for Tech's student newspaper.
- Increased Lighthouse Accessibility and SEO to **96/100** by rebuilding the UI/UX with TailwindCSS and React.
- **Reduced API load by 55%** by adding middleware that caches WordPress REST calls and merges frequent fetches.

PROJECTS

Cimic – AI ATL Hackathon Winner | React, Rust, Azure, LangChain, Python, Pandas, Tailwind

Nov. 2025

- Delivered a **245 neighborhood policy simulation** by normalizing GeoJSON census data with Python & Pandas.
- Improved policy-run efficiency by building a two-phase **Rust+LangChain** pipeline with **Azure AI** that used neighborhood context to pick affected areas and generate events, decreasing token count by **70%**.
- Enabled constituent feedback, measured by **30+ voices**, integrating **ElevenLabs** agents with contextual memory.
- Visualized simulated outcomes by creating an animated **React-Leaflet** impact map with live metric deltas.

JPMorganChase Code for Good Hackathon | TypeScript, Next.js, Docker, OpenAI, Twilio, AWS, PostgreSQL

Sept. 2025

- Built an **AI-driven SMS agent** using Mastra & Twilio to reducing manual event registration and discovery by **60%**.
- Prototyped a unified Next.js dashboard via Docker on AWS **EC2** for event creation, approvals, and analytics.
- Collaborated in a team of 7 and presented the solution to JPMorgan engineers and nonprofit judges.

Personality Academy | JavaScript, Next.JS, Prisma, PostgreSQL, CSS, ImageKit, GCP

Sept. 2022 - Aug. 2025

- Launched full-stack learning platform for **3,000+ users** across 23 chapters, tracked via completion analytics.
- Delivered sub-second **multi-filter search** for **2,000+ profiles**, grouping 512 categories via optimized SQL joins.
- Built management for **2,000 profiles + 13,000 content items**, replacing spreadsheets and saving **10+ hrs/week**.
- Cut image load times from **12s to 1s** with ImageKit CDN, sustaining **99.9% uptime** during traffic spikes.

TECHNICAL SKILLS

Languages: Python, JavaScript/TypeScript, Java, SQL, Go, Swift, HTML/CSS, C++, Assembly

Frameworks: React, Next.js, Node.js, Flask, FastAPI, Django, React Native, Express, TailwindCSS

AI/ML & Data: PyTorch, TensorFlow, Pandas, NumPy, Matplotlib, LangChain, RAG, Mastra

Dev & Tech: Git, Docker, AWS, Google Cloud, Azure, Supabase, Firebase, MongoDB, Vercel